(1) Prove the following statements using principle of Mathematical Induction for all $n \in N$.

a) n^2+3n is divisible by 2. b) $\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\dots\dots\dots+\frac{1}{n\times(n-1)}=\frac{n}{n+1}$

Solution : a) Let $P(n)$: n^2+3n is divisible by 2.

To prove that $P(1)$ is true.

$(1)^2+3(1)=1+3=4$, which is divisible by 2.

$\therefore$ $P(1)$ is true.

Now assume that $P(k)$ is true for some k.

i.e. assume that k^2+3k is divisible by 2.

i.e. assume that $k^2+3k=2m$ for some integer m.

To prove that $P(k+1)$ is true.

i.e. to prove that $(k+1)^2+3(k+1)$ is divisible by 2.

$$(k+1)^2+3(k+1)=k^2+2k+1+3k+3$$
$$=k^2+5k+4$$
$$=k^2+3k+2k+4$$
$$=2m+2k+4$$
$$=2(m+k+2),$$ where $m+k+2$ is an integer.

$\therefore$ $(k+1)^2+3(k+1)$ is divisible by 2.

$\therefore$ $P(k+1)$ is true whenever $P(k)$ is true.

By principle of mathematical induction, $P(n)$ is true for all $n\in N$.

b) Let $P(n)$: $\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\dots\dots\dots+\frac{1}{n\times(n-1)}=\frac{n}{n+1}$

Step I :- Let $n=1$

LHS $=\frac{1}{1\times(2)}=\frac{1}{2}$ RHS $=\frac{1}{1+1}=\frac{1}{2}$

LHS = RHS $\therefore$ P(1) is true

Step 2 :- let P(K) be true

$$\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\cdots+\frac{1}{K(K+1)}=\frac{K}{K+1} \quad \text{--- (1)}$$

Now we have to prove that P(K+1) is true

$$\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\cdots+\frac{1}{K(K+1)}+\frac{1}{(K+1)(K+2)}=\frac{K+1}{K+2}$$

$$LHS = \frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\cdots+\frac{1}{K(K+1)}+\frac{1}{(K+1)(K+2)}$$

$$= \frac{K(K+2)+1}{(K+1)(K+2)}$$

$$= \frac{K^2+2K+1}{(K+1)(K+2)}$$

$$= \frac{(K+1)^2}{(K+1)(K+2)} = \frac{(K+1)}{(K+2)}$$

= RHS

P(K+1) is true from step (1) & setp (2) & the principal of mathematical Induction P(n) is true $\forall n \in N$

2) Using Binomial Theorem expand :

a) $(x+3)^5$ b) $(x^2+6)^6$

Solution : We know that $(a+b)^n = {}^nC_0a^n + {}^nC_1a^{n-1}b + {}^nC_2a^{n-2}b^2 + \ldots\ldots + {}^nC_nb^n$

a) $(x+3)^5 = {}^5C_0x^5 3^0 + {}^5C_1x^4 3^1 + {}^5C_2x^3 3^2 + {}^5C_3x^2 3^2 + {}^5C_4x^2 3^4 + {}^5C_5x^0 3^5$

$= 1x^5 \cdot 1 + 5x^4(3) + \frac{5!}{3!2!}x^3(9) + \frac{5!}{3!2!}x^2(27) + \frac{5!}{4!1!}x(81) + \frac{5!}{5!0}\times 243$

$= x^5 + 15x^4 + \frac{5\times 4}{2}3x^3 + \frac{5\times 4}{2}27x^2 + 15\times 81x + 243$

$(x+3)^5 = x^5 + 15x^4 + 90x^3 + 270^2 + 405 + 243$

$+ {}^6C_0 (x^2)^6 (6)^0 + {}^6C_1 (x^2)^5 6^1 + {}^6C_2 (x^2)^4 6^2 + {}^6C_3 (x^2)^3 6^3$

$+ {}^6C_4 (x^2)^2 6^4 + {}^6C_5 (x^2)^1 6^5 + {}^6C_6 (x^2)^0 6^6$

$= x^{12} + 36x^{10} + 15x^8 36 + 20x^6 (216) + 15x^4 (1296) + 6x^2 (7776) + (1)(46656)$

$= x^{12} + 36x^{10} + 540x^8 + 4320x^6 + 19440x^4 + 46656x^2 + 46656$

3) Find the 5th term in the Binomial expansion of :

(a) $(x+3y)^7$ (b) $(2x-7)^4$

Solution : We know that $(r+1)^{th}$ term in the expansion of $(a+b)^n$ is $t_{r+1} = {}^nC_r a^{n-r} b^r$

a) $(x+3y)^7$

Put $x = 4$, $n = 7$, $a = x$

$b = 3y$

$t_5 = {}^7C_4 x^3 (3y)^4$

$= \frac{7!}{4!3!} x^3 (81y^4)$

$= \frac{7 \times 6 \times 5 \times 4!}{4! \times 3 \times 2 \times 1} \times 81y^4 \times x^3$

$\boxed{t_5 = 2835 x^3 y^4}$

b) $(2x-7)^4$

Put $x = 4$; $n = 4$, $a = 2x$, $b = -7$

$t_5 = {}^4C_4 (2x)^{4-4} (-7)^4$

$t_5 = 1 \times (2x)^0 (7)^4$ (7 is even)

$= 1 \times 1 \times 7^4$

$t_5 = 2401$

$\therefore \boxed{t_5 = 2401}$

4) Find the coefficient A of the given term in each of the binomial expansion :

Solution : We know that $(r+1)^{th}$ term in the expansion of $(a+b)^n$ is $t_{r+1} = {}^nC_r a^{n-r} b^r$

	Binomial	Term
a	$(x+5)^7$	$A x^6$
b	$(3y^{-2}-1)^8$	$A y^{-8}$
c	$\left(x-\frac{4}{x}\right)^6$	$A x^0$
d	$(x+2y)^4$	$A x^2 y^2$

a) Let the term in x^6 be $(r+1)^{th}$ term in $(x+5)^7$

$t_{r+1} = {}^nC_r a^{n-r} b^r = {}^7C_r x^{7-r} 5^r$ is term in x^6.

$\therefore\ 7-r=6$

$\therefore\ r=1$

$\therefore\ r+1=2$ and 2nd term is in x^6

$\therefore\ t_2 = {}^7C_1 (x)^6 (5)^1 = 35x^6$

The coefficient of x^6 is A=35

b) $(3y^2-1)^8$

Let the term in any y^{-8} be $(r+1)^{th}$ term in $(3y^{-2}-1)^8$

$t_{r+1} = {}^nC_r a^{n-r} b^r = {}^8C_r (3y^{-2})^{8-r} (-1)^r$ is term in y^{-8}

$-2(8-r) = -8$ i.e. $(8-r) = \frac{-8}{-2} \Rightarrow 8-r=4$ $\boxed{r=4}$

$r+1=5$ and 5th term is in y^{-8}

$t_5 = {}^8C_4 (3y^{-2})^4 (-1)^4$

$= \frac{8}{4!4!} \times 81 \times y^{-8}$

$= \frac{8\times7\times6\times5\times8^1}{4\times3\times2\times1} \cdot y^{-8} = 5670\, y^{-8}$

the coefficient of y^{-8} is $\underline{A = 5670}$

c) $\left(x-\frac{4}{x}\right)^6$

Let the term in x^0 be $(r+1)^{th}$ term in $\left(x-\frac{4}{x}\right)^6$

$t_{r+1} = {}^nC_r a^{n-r} b^r = {}^6C_r x^{6-r} \left(\frac{-4}{x}\right)^r$ is term in x^0

$\therefore\ 6-r=0 \Rightarrow \boxed{r=6} \Rightarrow r+1=7$ & 7th term in x^0

$t_7 = {}^6C_6 x^0 \left(\frac{-4}{x}\right)^6 = 1\times1\times\frac{4096}{x^6}$

$t_7 = \frac{4096}{x^6} \Rightarrow$ Coeff. of x^6 is $A = \underline{\frac{4096}{x^6}}$

d) $(x+2y)^4$

Let the term in the x^2y^4 be $(r+1)^{th}$ term in $(x+2y)^4$

$t_{r+1} = {}^nC_r\, a^{n-r} b^r \Rightarrow t_{r+1} = {}^4C_r\, x^{4-r} (2y)^r$ is term in x^2y^2

$4-r = 2$ or $r = 2$

$r+1 = 3$ and 3rd term in x^2y^2

$t_3 = {}^4C_2\, x^2 (2y^2)$

$= 6x^2 \cdot 4y^2$

$= 24x^2y^2$

The coeff. of x^2y^2 is $\underline{A = 24}$

5) Use Binomial theorem to express each of the following in the form of $a + ib$:

(a) $(1+i)^5$ (b) $(3-4i)^4$ (c) $(2-7i)^6$

Solution :

a) $(1+i)^5 = {}^5C_0\, 1^5 i^0 + {}^5C_1\, 1^4 i^1 + {}^5C_2\, 1^3 i^2 + {}^5C_3\, 1^4 i^3 + {}^5C_4\, 1\, i^4 + {}^5C_5\, 1^0 i^5$

$i^2 = 1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$

$= 1 + 5i + 10(-1) + 10(-i) + 5(1) + 1(i)$

$= 1 + 5i - 10 - 10i + 5 + i$

$(1+i)^5 = 4 - 4i$

b) $(3-4i)^4 = {}^4C_0\, 3^4 (-4i)^0 + {}^4C_4\, 3^3 (-4i)^1 + {}^4C_2\, 3^2 (-4i)^2 + {}^4C_3\, 3^1 (-4i)^3 + {}^4C_4\, 3^0 (-4i)\, 1$

$i^2 = -1$, $i^3 = i$, $i^4 = i$

$(3-4i) = (1)(81)(1) + 4(27)(-4i) + 6(9)(16i^2) + 4(3)(-64i^3) + (1)(1)(256)\, i^4$

$= 81 - 432i + 864(-1) - 768(-i) + 256(1)$

$= 81 - 432i - 864 + 768i + 256$

$(3-4i) = -527 + 336i$

c) $(2-7i)^6 = {}^6C_0\, 2^6 (-7i)^0 + {}^6C_1\, 2^5 (-7i)^1 + {}^6C_2\, 2^4 (-7i)^2 + {}^6C_3\, 2^3 (-7i)^3 + {}^6C_4\, 2^2 (-7i)^4 + {}^6C_5\, 2^2 (-7i)^5 + {}^6C_6 (2)^0 (-7i)^6$

$= 1(64)1 + 6(32)(-7i) + 15(16)(49i^2) + 20(8)(-343i^2) + 15(4)(240i^4) + 6(2)(-1)(807i^5) + (1)(1)(117649i^6)$

$= 64 - 1344i + 11760(-1) - 54880(-i) + 144060(1) - 201684(i) + 117649(-1)$

$= 64 - 1344i - 11760 + 54880i + 144060 - 201684i - 117649$

$(2-7i)^6 = 14715 - 148148i$

6) Using binomial theorem prove the following :

a) ${}^5C_0 + 2\,{}^5C_1 + 4\,{}^5C_2 + 8\,{}^5C_3 + 16\,{}^5C_4 + 32\,{}^5C_5 = 243$

b) ${}^{20}C_{20} - {}^{20}C_{19} + {}^{20}C_{18} - {}^{20}C_{17} + \ldots + {}^{20}C_2 - {}^{20}C_1 + {}^{20}C_0 = 0$

Solution :

(a) ${}^5C_0 + 2\,{}^5C_1 + 4\,{}^5C_2 + 8\,{}^5C_3 + 16\,{}^5C_4 + 32\,{}^5C_5 = 243$

$L.H.S. = {}^5C_0 + 2\,{}^5C_1 + 4\,{}^5C_2 + 8\,{}^5C_3 + 16\,{}^5C_4 + 32\,{}^5C_5$

$= 1 + 2\times5 + 4\times10 + 8\times10 + 16\times5 + 32\times1$

$= 1 + 10 + 40 + 80 + 80 + 32$

$= 243$

(b) ${}^{20}C_{20} - {}^{20}C_{19} + {}^{20}C_{18} - {}^{20}C_{17} + \ldots + {}^{20}C_2 - {}^{20}C_1 + {}^{20}C_0 = 0$

${}^nC_r = {}^nC_{n-r}$

$\therefore {}^{20}C_{20} = {}^{20}C_0,\ {}^{20}C_{19} = {}^{20}C_1,\ {}^{20}C_{18} = {}^{20}C_2$

$L.H.S. = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots + {}^{20}C_3 - {}^{20}C_2 + {}^{20}C_1 - {}^{20}C_0$

$L.H.S. = 0$

$\therefore L.H.S. = R.H.S.$